HIGHER-ORDER DIFFERENTIAL EQUATIONS | Unified Quick Reference

Engineering Mathematics • Differential Operators ↔ Complementary Function ↔ Particular Integral

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1 GENERAL FORM AND CLASSIFICATION

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = r(x)$$

Constant coefficients (a_k constants):

$$a_n y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1 y' + a_0 y = r(x)$$

- **Order**: highest derivative appearing.
- **Linear**: y and its derivatives occur only to the first power, with no products.
- **Homogeneous**: $r(x) = 0$; **non-homogeneous**: $r(x) \neq 0$.

General solution $y = y_c + y_p$, where y_c = Complementary Function (CF) and y_p = Particular Integral (PI).

CF y_c solves $F(D)y = 0$, n constants + **PI** y_p any one solution of $F(D)y = r$ = **General Solution** $y_c + y_p$

General Solution = CF + PI

2 DIFFERENTIAL OPERATOR METHOD

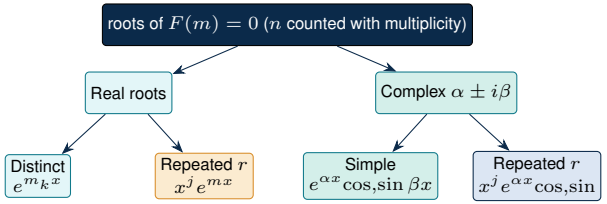
$$D = \frac{d}{dx}, \quad D^n y = \frac{d^n y}{dx^n}$$

$$F(D) = a_n D^n + \cdots + a_1 D + a_0, \qquad F(D)y = r(x)$$

Homogeneous: $F(D)y = 0$. Characteristic equation: $\boxed{F(m) = 0}$

Trying $y = e^{mx}$ gives $F(D)e^{mx} = F(m)e^{mx}$, so e^{mx} solves $F(D)y = 0$ exactly when $F(m) = 0$. The roots of $F(m) = 0$ (counted with multiplicity) determine the complementary function. Since $F(D) = a_n(D - m_1)\cdots(D - m_n)$ has constant coefficients, its factors commute.

3b ROOT CLASSIFICATION DIAGRAM



3 COMPLEMENTARY FUNCTION: ROOT CASES

1. Distinct real $m_1, \ldots, m_n$	$y_c = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \cdots + C_n e^{m_n x}$
2. Repeated real m of multiplicity r	$y_c = e^{mx}(C_1 + C_2 x + \cdots + C_r x^{r-1})$
3. Complex pair $m = \alpha \pm i\beta$	$y_c = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$
4. Repeated complex $\alpha \pm i\beta$, multiplicity r	$y_c = e^{\alpha x} \sum_{k=0}^{r-1} x^k \times (C_{2k+1} \cos \beta x + C_{2k+2} \sin \beta x)$

The number of independent constants equals the order n. A repeated factor $(D - m)^r$ contributes r solutions $e^{mx}, xe^{mx}, \ldots, x^{r-1}e^{mx}$. Complex roots of real-coefficient equations occur in conjugate pairs. Colour code: **real/complex simple**, **repeated real**, **repeated complex**.

4 PARTICULAR INTEGRAL: OPERATOR FORMULAS

$$y_p = \frac{1}{F(D)} r(x)$$ (superposition: if $r = r_1 + r_2$, then $y_p = y_{p1} + y_{p2}$)

A. Exponential, $r = e^{ax}$: $\boxed{\frac{1}{F(D)} e^{ax} = \frac{e^{ax}}{F(a)}}$, $F(a) \neq 0$

B. Polynomial, $r = x^n$, with $F(0) \neq 0$: put $\varphi(D) = \dfrac{F(D) - F(0)}{F(0)}$,

$$\frac{1}{F(D)} x^n = \frac{1}{F(0)} \left[1 - \varphi(D) + \varphi(D)^2 - \cdots\right] x^n$$

Valid only for $F(0) \neq 0$. Since $D^{n+1}x^n = 0$, terms beyond D^n are dropped and the series terminates. If $F(0) = 0$, write $F(D) = D^k G(D)$ with $G(0) \neq 0$ and integrate k times after applying $1/G(D)$.

C. Sine and cosine, $r = \sin ax$ or $\cos ax$: use e^{iax} and take Im or Re.

If $F(ia) \neq 0$: $\dfrac{1}{F(D)} e^{iax} = \dfrac{e^{iax}}{F(ia)}$

So $\dfrac{1}{F(D)} \sin ax = \text{Im}\, \dfrac{e^{iax}}{F(ia)}$, $\dfrac{1}{F(D)} \cos ax = \text{Re}\, \dfrac{e^{iax}}{F(ia)}$

Shortcut when $F(D) = \phi(D^2)$: $\dfrac{\sin ax}{\phi(-a^2)}$, $\phi(-a^2) \neq 0$.

Shift rule: $\dfrac{1}{F(D)} \left[e^{ax} V(x)\right] = e^{ax} \dfrac{1}{F(D + a)} V(x)$

5 RESONANCE AND MODIFIED PARTICULAR INTEGRAL

Warning: if $F(a) = 0$ the formula $e^{ax}/F(a)$ fails

For a root a of multiplicity s: $F(D) = (D - a)^s G(D)$, $G(a) \neq 0$

For $r(x) = e^{ax}$: $\boxed{y_p = \dfrac{x^s e^{ax}}{s!\, G(a)}}$

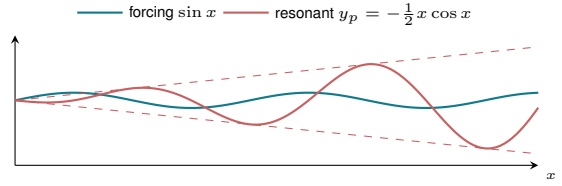
Simple root ($F(a) = 0$, $F'(a) \neq 0$): $\boxed{y_p = \dfrac{xe^{ax}}{F'(a)}}$

Trigonometric case: if $F(ia) = 0$ with $F(D) = (D - ia)^s G(D)$, $G(ia) \neq 0$, then $y_p = \text{Im}$ or Re of $\dfrac{x^s e^{iax}}{s!\, G(ia)}$.

Non-resonance $F(a) \neq 0$ → Standard PI $e^{ax}/F(a)$

Resonance $F(a) = 0$ → Multiply by x^s (sufficient power)

The modified PI must be linearly independent of the CF. Physically, forcing at a natural frequency makes the amplitude grow like x^s (plot: $y'' + y = \sin x$ has $y_p = -\frac{1}{2} x \cos x$).



⋆ WORKED CHECK (CF + PI)

Solve $y'' - 3y' + 2y = e^x$. CF: $m^2 - 3m + 2 = 0 \Rightarrow m = 1, 2$, $y_c = C_1 e^x + C_2 e^{2x}$. Here $a = 1$ is a simple root ($F'(m) = 2m - 3$, $F'(1) = -1$), so $y_p = \dfrac{xe^x}{F'(1)} = -xe^x$.

$\boxed{y = C_1 e^x + C_2 e^{2x} - xe^x}$

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6 METHOD OF UNDETERMINED COEFFICIENTS

For constant coefficients and forcing built from constants, polynomials, e^{ax}, sin, cos and their products.

Forcing function	Trial particular integral
Constant K	A
Polynomial $P_n(x)$	general polynomial of degree n
e^{ax}	Ae^{ax}
$\sin ax$ or $\cos ax$	$A\cos ax + B\sin ax$
$e^{ax}P_n(x)$	$e^{ax}\times$ (general polynomial of degree n)
$e^{ax}\sin bx$ or $e^{ax}\cos bx$	$e^{ax}(A\cos bx + B\sin bx)$

Resonance rule: if any term of the trial overlaps the CF, multiply the *entire* trial by x^s, where s is the multiplicity of the overlapping root.

Example. $y'' - 3y' + 2y = e^x$: CF has e^x, so the trial Ae^x fails; use $y_p = Axe^x$. Substituting gives $-A = 1$, so $A = -1$.

7 VARIATION OF PARAMETERS

For $y'' + P(x)y' + Q(x)y = R(x)$ (leading coefficient *normalised to 1*), let y_1, y_2 be linearly independent solutions of the homogeneous equation.

$$W = y_1y_2' - y_1'y_2$$

$$\boxed{y_p = -y_1\int \frac{y_2R}{W}\,dx + y_2\int \frac{y_1R}{W}\,dx}$$

- Works for any continuous $R(x)$ (e.g. $\sec x$, $\ln x$, $1/x$) and for variable coefficients.
- Needs y_1, y_2 linearly independent: $W \neq 0$ on the interval.

Find CF → Compute Wronskian → Evaluate integrals → Obtain PI

8 REDUCTION OF ORDER

For $y'' + P(x)y' + Q(x)y = 0$ with one known solution $y_1 \neq 0$, put $y_2 = y_1v$:

$$\boxed{y_2 = y_1\int \frac{e^{-\int P(x)\,dx}}{y_1^2}\,dx}$$

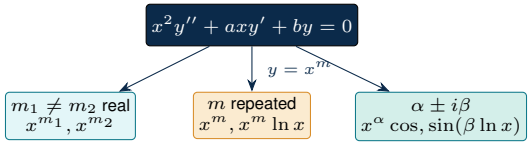
Reduction of order converts a second-order homogeneous equation into a first-order equation for the auxiliary function v'. (Normalised form with leading coefficient 1; use on intervals where $y_1 \neq 0$.)

9 CAUCHY–EULER EQUATIONS

$x^2y'' + axy' + by = r(x)$. Homogeneous case: assume $y = x^m$, giving the auxiliary equation $\boxed{m(m-1) + am + b = 0}$

Distinct real m_1, m_2	$y_c = C_1x^{m_1} + C_2x^{m_2}$
Repeated real m	$y_c = x^m(C_1 + C_2\ln x)$
Complex $\alpha \pm i\beta$	$y_c = x^\alpha\big(C_1\cos(\beta\ln x) + C_2\sin(\beta\ln x)\big)$

Domain: formulas are for $x > 0$; for $x < 0$ replace x by $|x|$ ($x = 0$ is a singular point). **Substitution** ($x > 0$): $t = \ln x$, $\theta = d/dt$ gives $xy' = \theta y$, $x^2y'' = \theta(\theta-1)y$, so $\dfrac{d^2y}{dt^2} + (a-1)\dfrac{dy}{dt} + by = r(e^t)$ (constant coefficients).



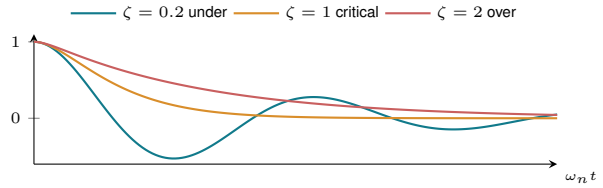
10 ENGINEERING APPLICATION: MECHANICAL VIBRATIONS

$$\boxed{m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = F(t)}$$

m mass, c damping coefficient, k stiffness, $F(t)$ external force, $x(t)$ displacement.

$\omega_n = \sqrt{k/m}$, $\quad \zeta = \dfrac{c}{2\sqrt{mk}}$ (so $\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = F/m$)

$\zeta < 1$ Underdamped	$x_h = e^{-\zeta\omega_nt}(C_1\cos\omega_dt + C_2\sin\omega_dt)$, $\omega_d = \omega_n\sqrt{1-\zeta^2}$
$\zeta = 1$ Critical	$x_h = e^{-\omega_nt}(C_1 + C_2t)$
$\zeta > 1$ Overdamped	$x_h = C_1e^{\lambda_1t} + C_2e^{\lambda_2t}$, $\lambda_{1,2} = -\omega_n\big(\zeta \pm \sqrt{\zeta^2-1}\big)$



Free response ($F = 0$): x_h above, with $x(0) = 1$, $\dot{x}(0) = 0$ in the plot. **Forced response:** $x = x_h + x_p$. For $\zeta > 0$, $x_h \to 0$ (transient) and x_p remains. For $F = F_0\cos\omega t$, $c > 0$: amplitude $X = \dfrac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$. If $c = 0$ and $\omega = \omega_n$, resonance: $x_p \propto t$ grows.

11 INITIAL CONDITIONS AND SOLUTION CONSTANTS

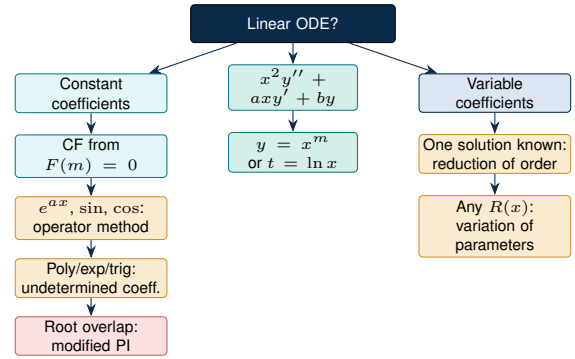
For $y'' + P(x)y' + Q(x)y = R(x)$ with $y(x_0) = y_0$, $y'(x_0) = v_0$:

- The general solution has **two** independent constants; the initial conditions fix them.
- For an nth-order equation, n independent conditions are normally needed. Uniqueness holds under existence–uniqueness conditions: continuous coefficients and R on an interval containing x_0, with leading coefficient $\neq 0$.

Workflow: 1. identify order and type. **2.** find CF. **3.** find PI. **4.** form $y = y_c + y_p$. **5.** apply initial or boundary conditions. **6.** verify by substitution.

12 METHOD SELECTION GUIDE

Equation type	Recommended method
Constant coefficients, homogeneous	Characteristic equation
Constant coefficients, exponential forcing	Operator method
Polynomial, exponential, sine/cosine forcing	Undetermined coefficients
Resonant forcing	Modified PI ($\times x^s$)
General second-order non-homogeneous	Variation of parameters
One known solution of a homogeneous equation	Reduction of order
Cauchy–Euler equation	$y = x^m$ or $t = \ln x$
Mechanical vibration model	Characteristic roots + forced response



13 COMMON ERRORS: QUICK CHECK

- Forgetting the multiplicity of a repeated characteristic root.
- Omitting the x-factor in resonance.
- Selecting an incomplete trial PI.
- Confusing complementary function and particular integral.
- Applying undetermined coefficients to unsuitable forcing functions.
- Forgetting the Wronskian in variation of parameters.
- Applying Cauchy–Euler formulas without considering the domain.
- Forgetting to apply initial conditions.
- Failing to verify the final solution by substitution.